

This is an account of the problems met while building an offline Hindi voice assistant on a single 8 GB board, and of what each one cost to find. Three of them changed a model. Several of them were only visible because something was measured that did not have to be measured — and two of them were mistakes in our own measurements rather than in the device. Every number here was taken on this board.

Models we changed, and why

four changes

Hindi speech recognition was dropping nine consonants of the alphabet

PROBLEM Words came back missing letters, and always the same letters: मुझे as मुे, कुछ as कु, समझ as सम. Clean audio at the right sample rate made no difference.

CAUSE The ONNX export we started from carried a hardcoded 89-token vocabulary. The real model has 257. Nine Devanagari consonants were absent from the table altogether (झ छ घ ठ ढ ण ष ड ञ), and because the CTC blank index is computed as the vocabulary length, every one of the model's other output classes above index 88 was silently discarded too.

FIX Traced the checkpoint back to AI4Bharat's indic-conformer-600m-multilingual and switched to a public int8-quantised export of **the same model** that ships the complete 257-token Hindi vocabulary and its language mask. The model family did not change — the export did.

RESULT Round-trip synthesis→recognition on the previously failing sentences now returns character-for-character matches.

The language model reversed a clinical negation

PROBLEM With the rule “Do not bathe a baby until its weight is 2000 gm” present word for word in the retrieved page, the model answered that the baby should be bathed daily until its weight reaches 2000 grams. On a health device that is not a weak answer; it is a dangerous one.

CAUSE We had moved to qwen2.5:1.5b purely to save memory. It was not merely weaker at this size — it was unsafe on negation.

FIX Measured four candidates on the same question and chose on behaviour, not size:

MODEL	RESIDENT	BEHAVIOUR ON THE NEGATION TEST
qwen2.5:1.5b	817 MB	negation reversed — unsafe
qwen2.5:3b	1684 MB	“I do not have this information” — safe but wrong
qwen3-vl:2b	3084 MB	correct, but rambles and will not fit
qwen3:1.7b	921 MB	correct and concise — chosen

qwen3:1.7b is the text-only sibling of the vision model we had been running. The camera-less application could never use that model's CLIP tower, which was costing a 322 MB reserved vision buffer and 721 extra tensors at every start. Dropping it freed **2.2 GB** — the difference between the board swap-thrashing after about six questions, with available memory down to 60 MB and audio playback stalling in uninterruptible sleep, and a steady state of 34 MB of swap.

The Hindi voice had to be chosen against the clock, not by ear

PROBLEM Synthesis was the single largest stage in the pipeline and the voice we liked best could not run in the time budget.

FIX All three candidates were run on the same sentences through this device's own speaker:

VOICE	SYNTHESIS	WEIGHTS	RATE	LICENCE
MMS VITS	~1.0 s	250 MB	16 kHz	CC-BY-NC
Piper	0.48 s	64 MB	22 kHz	CC-BY-NC-SA
IndicF5	5.8–10.4 s	1.4 GB	24 kHz	MIT

IndicF5 sounded clearly best and has the cleanest licence, so it was pursued properly before being ruled out: fp16 diverges to NaN under both half precision and autocast, reducing the solver steps below eight degrades audibly, and trimming the reference clip from 8 s to 3 s did give a threefold speedup. Even then the floor was 5.8 s for a six-word answer, against a 5–8 s budget for the whole pipeline. It is the right upgrade on a larger board, not on this one. Piper's Hindi voice is trained on AI4Bharat's indicnlp corpus, so the Indic provenance is preserved. Its licence is non-commercial and would have to change before sale.

The translator was cutting the end off spoken answers

PROBLEM A forty-word answer stopped partway through the instruction: “...that requires immediate intervention” was spoken as “... जिसके लिए तत्काल हस्तक्षेप”, with the verb missing.

CAUSE The decoder’s maximum output length was 64 tokens. Hindi needs more tokens than the English it is translated from, so the ceiling was reached mid-sentence.

FIX Raised to 192. With greedy decoding this costs nothing unless the output actually runs long. IndicTrans2 itself was never replaced.

The board itself

five issues

The board’s button cannot report that it is being held

PROBLEM Every question longer than about eight seconds was cut off. This was worse than it sounds: “... डॉक्टर या नर्स मौजूद नहीं है” lost its negation to the cut, the translator rendered the remains as the doctor is present, and the answer inverted with it. Several reports of “wrong answer” traced back to this one pin.

CAUSE Measured repeatedly, with raw polling and with edge callbacks, both agreeing: the trigger pin goes high on press and drops low after about 7.8 s (7.76 / 7.78 / 7.79 / 7.81 / 7.90 / 7.90 / 7.91 s across runs) while the button is still physically held, and never returns high. The recorder itself runs cleanly past eighteen seconds, so this is the pin, not the software. Jetson.GPIO’s pull-up request is ignored on this platform; changing it for real needs a device-tree pinmux rebuild, and a pull-down would not fix a high level that decays anyway.

FIX The interaction was redesigned around the hardware rather than fought: press once, speak, stop talking. Recording ends after 3.5 s of silence, with the threshold calibrated from the room’s own noise floor at the start of each recording. Pressing again still stops it immediately. 3.5 s and not 2.0 s, because at 2.0 s an ordinary breath between sentences ended the question and the fragment that reached the model contained no question at all.

The first question of every session was the slowest one

PROBLEM Recognition took 6–8 s on question one and about 0.65 s afterwards. The very first thing a user experiences was the worst thing they experience.

CAUSE The warm-up exercised the ONNX sessions directly, so the first real request still paid the audio library’s just-in-time compilation. The translator had no warm-up at all, and warming it inside its own constructor was not enough — synthesis builds another runtime session afterwards, so the warm-up has to run once every engine is up.

FIX Warm every engine through the same entry point real requests use, after all of them are loaded, and with a full sentence rather than a two-word greeting, because kernels are selected for the shape actually given.

RESULT First question 0.66 s. Translation 0.51 s on the first request and on every later one.

Long questions degraded into nonsense rather than failing cleanly

PROBLEM The recogniser is trained on short utterances. Measured against known text: 11.0 s kept 95% of the characters, 20.2 s kept 70%, 28.6 s 52%, 38.2 s 42% — and the tail turned to nonsense instead of stopping.

CAUSE Attention, not a shape limit: every axis in the ONNX graph is dynamic.

FIX Long audio is split near twelve-second boundaries, each boundary nudged to the quietest point within 1.5 s so the seams fall in word gaps, and each chunk is normalised and decoded on its own.

RESULT 95% at every length tested, in 1.3–1.9 s.

Devanagari could not be drawn on the screen at all

PROBLEM The display library blits one bitmap glyph per code point with no shaping. Matras floated detached above the line, the pre-base i-matra was never reordered (मासिक drew as मासक), and conjuncts never formed (स्तनपान drew as सतनपान).

FIX Text is shaped with HarfBuzz into a one-bit image and pushed into the display buffer — correct for both scripts, about 33 ms per update. The screen also used to flip between languages mid-interaction, showing the English translation and then the English answer before restoring Hindi; those were pipeline internals leaking into the interface, and a worker who asks in Hindi now sees only Hindi.

Power mode, and where the remaining latency went

PROBLEM Generation was the dominant cost and the board was not being asked for everything it has.

FIX MAXN_SUPER, made persistent across reboot; and the answer is now spoken sentence by sentence as it is generated rather than after it is complete.

RESULT Generation 35.8 tokens/s, steady over ten runs at 52 °C. Language model 4.34 s → 2.51 s. First audio 5.37 s → 4.05 s. Synthesis moved to the GPU separately: 4.3 s → 0.9 s.

The manual arrived as a PDF that had been printed backwards

PROBLEM Text extracted from the source PDF was corrupt in four distinct ways at once: letters doubled inside words, running heads reversed, whole pages reversed character by character (mroF tisiV emoH for Home Visit Form), and bullet markers turned into the letter u.

FIX The whole extracted book was audited page by page and repaired on load: 82 doubled words, 78 reversed heads, 4 fully reversed pages and 35 bullets. Of the source PDF's 88 pages, 12 yield no text at all — they are images or blank.

RESULT 72 pages are indexed. Four of the reversed pages are the home-visit checklists; they were deliberately left out of the index after measurement, because including them pushed the clean list of sepsis danger signs out of the retrieved context and the evaluation fell from 9/10 to 8/10.

Where the real accuracy problem was hiding

PROBLEM On a 52-question set — one question from every page of the manual from page 20 onward — only 42 were answered and only 30 came from the page the question was written against.

CAUSE Not the ranking. With the relevance gate switched off, the correct page was already in the top three for 71% of them. It was the vocabulary: the question is asked in Hindi and the book is written in English, and the translator kept choosing the everyday sense of a word the manual uses clinically. Question and answer then shared no word at all.

FIX Eight substitutions, each chosen by translating the real question both ways and keeping the wording that produced the term the index is actually built on.

WHAT SHE SAYS	WHAT THE TRANSLATOR PRODUCED	WHAT THE MANUAL PRINTS
आशा	the role of hope	ASHA
पानी की कमी	water scarcity, no water	dehydration — on all four pages that answer it
यौन संक्रमण	sexual transition	sexually transmitted infection
दम घुटना	choking	asphyxia
छल्ला	cut open	IUCD
नसबंदी	vasectomy — even for a woman	sterilisation
पसली चलना	walk rib	chest indrawing

RESULT Answered 42 → 50 of 52; answered from the page the question came from 30 → 39. One of these was not cosmetic: asked what medicine an ASHA may give for a breathing illness, the device had been answering chloroquine or ACT — malaria drugs — for pneumonia. The words ASHA and worker appear on nearly every page of a manual written for her, so they carried a page of session objectives and two TB pages above the classification table that names the drug. Dropped from both the question and the index, the same question now answers **Cotrimoxazole**.

The recogniser does not give the same answer twice

PROBLEM One pass of the 52 questions said 45 keywords survived; another said 50. Neither was a measurement.

CAUSE The recogniser is not deterministic. The same question, spoken the same way, came back as फसली, पसी and फसली across three tries.

FIX Everything about speech is now measured over 156 recognitions — all 52 questions, three times each — not one pass. Two narrow repairs were added on that evidence: a dropped letter costs a whole point of edit distance and could never clear the confusable-slip limit, so स्तनपान heard as स्नपान went through untouched; and two nasal marks written next to each other spell no Hindi word but, being two different characters, walked past the rule that collapses repeats.

RESULT 97% of keywords survived before the repairs, 99% after.

A rule that was fixing words was also destroying one

PROBLEM The device answered “what can an ASHA do **to** a pregnant woman” instead of “to help a pregnant woman”.

CAUSE A rule that collapses stuttered letters was collapsing मदद — help — into मद, and with the help gone from the sentence the translator dropped it from the English too.

FIX A doubled vowel sign or nasal is never a real Hindi spelling and is still always collapsed. A doubled consonant can be real, so it is now collapsed only where doing so recovers one of the clinical terms. Over the 156 recognitions the rule fires on five distinct tokens: it repairs three of them and no longer breaks मदद.

The device gave contraceptive advice to a pregnant woman

PROBLEM Asked what a pregnant woman with anaemia should watch for, the device said she should avoid an IUCD. Asked what checkups she needs in her first months, that she needs a monthly pelvic exam to confirm the IUCD is in place. Asked which tablet prevents anaemia, that it is Mala N or Mala D — an oral contraceptive. None of it was hedged.

CAUSE This module is about children. It has ten pages on preventing pregnancy and none on antenatal care, so pregnant woman finds the contraception pages honestly. No threshold separates these: their relevance, 4.1 to 6.1 per term, sits inside the range of real questions, and coverage does not separate them either — 0.48 to 0.79, where real questions run from 0.39 up.

FIX The rule is stated as what it is. A page about how not to become pregnant cannot answer a question about a woman who already is, and is removed from the context; if nothing else was retrieved the device says it does not have the information, which for this manual is true. What the manual does cover about her — malaria in pregnancy, TB treatment in pregnancy, her tetanus shot — is untouched.

Answers that talked about themselves, and names that were translated away

PROBLEM Several faults that only appear when the answer is read aloud to someone who cannot see the screen. The model opened with YES and then said, correctly, that pills should not be advised — the contradiction first, which is the part she would act on. It said the context does not provide information, and the device dutifully translated the word context to a worker who has never been shown one. It repeated the manual's own teaching scaffolding back as an answer: Step 3 of the session includes... And the names of things she is handed were translated into the ordinary words they are made of: Gentian violet paint became जेंटियन बैंगनी रंग — a purple colour; the cord became डोर, a rope; zinc became जस्ता, the metal.

FIX Refusals are recognised whether the model uses the words it was given or its own. Bare verdicts and scaffolding are stripped — and stripped repeatedly, because the model stacks them ("The answer is: Step 3 of the session includes..." carries two). Names are steered from the English where that works and corrected on the Hindi where it does not, since how the translator renders a name depends on the sentence around it.

RESULT Checked against all fifty answered questions: exactly the answers that needed changing changed.

An ambiguous English word that named the wrong substance

PROBLEM Among the signs of dehydration the device read out खराब शराब पीना — drinking **alcohol** badly.

CAUSE English drink with no object is ambiguous, and the translator resolves it to alcohol often enough to matter.

FIX Give the verb an object. Probed the rest of the domain's ambiguous words the same way — stool, labour, discharge, delivery, passing, water breaking — and they all resolve correctly in context, so this stayed one narrow substitution rather than a glossary. It had to be fixed twice: the first fix caught the verb followed by an adverb (drinking poorly) but not the adjective in front of the noun (poor drinking), which is how the manual actually writes it.

What we got wrong about our own measurements

four issues

A crash that hid itself inside the one path the tests did not run

PROBLEM Every streamed answer became "this is not in the ASHA manual" while retrieval was scoring 4.75 and the right page was in hand.

CAUSE A variable was used in a worker thread that was never passed into it. The exception was swallowed and the failure looked exactly like a refusal. It went unnoticed because the evaluation harness never exercised the streaming path — the path the device actually uses.

FIX The variable is threaded through, and a failure now falls back to answering the ordinary way rather than telling a health worker the manual does not cover her question.

Two of our own measurements were wrong before the device was

PROBLEM A comparison of two prompt wordings appeared to show one clearly better. It was not.

CAUSE The test script truncated the retrieved context to 2400 characters while the device sends all of it, so the prompt under test was not the prompt in use. Separately, the evaluation harness kept its own copy of the prompt, which had drifted from the device's, and was scoring a wording the device no longer used.

FIX Both now import the prompt and the generation settings from the application, so a change to either is measured by the next run rather than going unnoticed.

Three changes that measured worse and were reverted

PROBLEM Not every good idea survived contact with the numbers.

FIX Splitting pages into paragraph-sized passages is the textbook improvement to this kind of retrieval. Measured: grounding fell from 0.67 to 0.63, strongly-grounded answers from 31 to 29, and one answer regressed from the right drug to the wrong one. Reverted. An early correction list for misheard words folded sounds together too aggressively and turned दूध into दस्त — milk into diarrhoea — damaging 6 of 19 good sentences; rebuilt from observed confusions only, it damages none of 28. And un-reversing the four checklist pages, which looked like an obvious repair, dropped the evaluation from 9/10 to 8/10.

RESULT Every change in this project is now kept or discarded on three checks: the 52-question page-by-page set, the ten-question evaluation, and twelve off-manual questions that must all be refused.

A document that reads correctly but cannot be copied

PROBLEM Text copied out of our own Hindi PDFs comes back broken — चाहिए copies as चा हए, यदि as द — and this was briefly mistaken for the device having answered differently. It had not.

CAUSE Wherever a vowel sign is reordered for display, the PDF loses the mapping back to the characters behind it. All eleven Devanagari fonts on this board were tried and every one loses it, most far worse than the one in use, so this is the renderer rather than the font.

FIX The test sheet is published as UTF-8 text alongside the PDF. The PDF is for reading and printing; the text file is for copying.

What is still open, stated plainly

The knowledge base is one module of seven. Module 7 covers child health and nutrition. Antenatal care, bleeding after delivery and anaemia in pregnancy are not in it, so questions about them have no right answer to find and the device is built to say so. Adding the remaining modules is the single largest improvement available, and it is a content task, not an algorithmic one.

Two answers in the 52 are wrong and are documented as wrong on the test sheet, along with two refusals and three loose answers. Both wrong answers are ranking failures on pages whose topic runs across a facing pair; neither is hidden.

Asking the question in Hindi against an English book is the remaining structural cost. Eight vocabulary substitutions were needed to close the gap by hand, and we cannot know which word a jury will use next. Indexing the Hindi edition of the same volume would remove that whole class of failure, and is the first thing to do after this stage.

Two licences must change before sale: the Hindi voice and the voice it replaced are both non-commercial.